C3 Solutions: differential calculus I

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All derivatives, extremes and tangent lines were checked with SymPy.

Practice

1a. $f(3) = 27 - 6 = 21$, $f(1) = 1 - 2 = -1$. Average rate of change $\dfrac{21 - (-1)}{3 - 1} = 11$.

1b. $f(2) = 4$. $f(2+h) = (2+h)^3 - 2(2+h) = 8 + 12h + 6h^2 + h^3 - 4 - 2h = 4 + 10h + 6h^2 + h^3$. $\dfrac{f(2+h) - f(2)}{h} = \dfrac{10h + 6h^2 + h^3}{h} = 10 + 6h + h^2$ for $h \neq 0$. As $h \to 0$ this tends to 10, so $f'(2) = 10$. (Check with the rules: $f'(x) = 3x^2 - 2$, $f'(2) = 10$.)

2a. $f(x+h) - f(x) = (x+h)^2 - 3(x+h) - x^2 + 3x = 2xh + h^2 - 3h$. Divide by h: $2x + h - 3 \to 2x - 3$. So $f'(x) = 2x - 3$.

2b. $\dfrac{1}{x+h+1} - \dfrac{1}{x+1} = \dfrac{(x+1) - (x+h+1)}{(x+h+1)(x+1)} = \dfrac{-h}{(x+h+1)(x+1)}$. Divide by h: $\dfrac{-1}{(x+h+1)(x+1)} \to \dfrac{-1}{(x+1)^2}$. So $f'(x) = -\dfrac{1}{(x+1)^2}$.

3a. $f'(x) = 20x^4 - 6x$.

3b. $6x^{1/2} \to 3x^{-1/2} = \dfrac{3}{\sqrt{x}}$.

3c. $2x^{-3} \to -6x^{-4} = -\dfrac{6}{x^4}$.

3d. Product rule: $f'(x) = 2x(x^3 - x) + (x^2 + 1)(3x^2 - 1) = 2x^4 - 2x^2 + 3x^4 + 2x^2 - 1 = 5x^4 - 1$. Expanding first: $f(x) = x^5 - x^3 + x^3 - x = x^5 - x$, so $f'(x) = 5x^4 - 1$. Same answer.

3e. $f'(x) = \dfrac{3(x+2) - (3x-1)\cdot 1}{(x+2)^2} = \dfrac{7}{(x+2)^2}$.

3f. $x^2\sqrt{x} = x^{5/2}$, so $f'(x) = \frac{5}{2}x^{3/2} = \frac{5}{2}x\sqrt{x}$.

4. $f(2) = 8 - 6 = 2$. $f'(x) = 3x^2 - 3$, $f'(2) = 9$. $y = 9x + b$ through $(2, 2)$: $b = 2 - 18 = -16$. **Tangent line:** $y = 9x - 16$.

5. $f'(x) = 6x^2 + 6x - 12 = 6(x+2)(x-1)$. Zeros $x = -2$ and $x = 1$. The sign of f' is $+, -, +$ (an upward parabola). Maximum $f(-2) = -16 + 12 + 24 = 20$. Minimum $f(1) = 2 + 3 - 12 = -7$.

6a. $f'(x) = 4x^3 - 16x = 4x(x-2)(x+2)$. Zeros $-2, 0, 2$, each simple; for large x, $f' > 0$. Signs: $-, +, -, +$. Decreasing on $(-\infty, -2]$ and $[0, 2]$; increasing on $[-2, 0]$ and $[2, \infty)$.

6b. Minima $f(-2) = f(2) = 16 - 32 = -16$. Maximum $f(0) = 0$.

7. $f'(x) = 4x^3 - 12x$, $f''(x) = 12x^2 - 12 = 12(x-1)(x+1)$. f'' changes sign at $x = -1$ (from $+$ to $-$) and at $x = 1$ (from $-$ to $+$). $f(\pm 1) = 1 - 6 + 1 = -4$. **Inflection points:** $(-1, -4)$ and $(1, -4)$.

8a. $v(t) = s'(t) = 6t^2 - 30t + 36$, $a(t) = v'(t) = 12t - 30$.

8b. $v(t) = 6(t^2 - 5t + 6) = 6(t-2)(t-3) = 0$: at $t = 2$ s and $t = 3$ s.

8c. $a(t) = 0 \iff t = 2.5$ s. Then $v(2.5) = 6 \cdot 0.5 \cdot (-0.5) = -1.5$ m/s. (This is the largest speed backwards: v has its minimum where $v' = a = 0$.)

9. Parallel means slope 4: $f'(x) = 2x = 4$, so $x = 2$, point $(2, 5)$. $y = 4x + b$ through $(2, 5)$: $b = -3$. **Tangent line:** $y = 4x - 3$.

CE-level

10a. After cutting, the base is a square of side $30 - 2x$ and the height is x. So $V = x(30 - 2x)^2$. We need $x > 0$ and $30 - 2x > 0$: domain $0 < x < 15$.

10b. Product rule: $V'(x) = 1 \cdot (30-2x)^2 + x \cdot (-4(30-2x))$. (For the derivative of $(30-2x)^2$, expand it to $900-120x+4x^2$: derivative $-120+8x = -4(30-2x)$.) So $V'(x) = (30-2x)((30-2x) - 4x) = (30-2x)(30-6x)$. $V'(x) = 0 \iff x = 15$ or $x = 5$; only $x = 5$ is in the domain. For $0 < x < 5$, both factors are positive: $V' > 0$. For $5 < x < 15$: $30 - 2x > 0$ and $30 - 6x < 0$, so $V' < 0$. Maximum at $x = 5$. **Maximum volume:** $V(5) = 5 \cdot 20^2 = 2000$ cm³.

11a. Quotient rule: $f'(x) = \dfrac{2x(x-1) - x^2 \cdot 1}{(x-1)^2} = \dfrac{x^2 - 2x}{(x-1)^2}$.

11b. $f'(x) = 0 \iff x(x-2) = 0 \iff x = 0$ or $x = 2$. The denominator is positive (for $x \neq 1$), so the sign of f' is that of $x(x-2)$: $+$ for $x < 0$, $-$ for $0 < x < 1$ and $1 < x < 2$, $+$ for $x > 2$. Maximum $f(0) = 0$; minimum $f(2) = \dfrac{4}{1} = 4$.

12a. $f_p'(x) = 3x^2 - p$. Two extreme values need two zeros of f_p' where the sign changes: $3x^2 = p$ has two solutions iff $p > 0$. (For $p = 0$, $f_0'(x) = 3x^2 \geq 0$ has one zero without a sign change; for $p < 0$ no zeros.) **Answer:** $p > 0$.

12b. At an extreme, $3x^2 = p$. Then $f_p(x) = x^3 - px = x^3 - 3x^2 \cdot x = -2x^3$. So the point $(x, f_p(x))$ lies on $y = -2x^3$. □

13. Let the touching point be (a, a^2). Slope $f'(a) = 2a$. Tangent: $y = 2a(x-a) + a^2 = 2ax - a^2$. It passes through $P(1, -3)$: $-3 = 2a - a^2$, so $a^2 - 2a - 3 = 0$, $(a-3)(a+1) = 0$, $a = 3$ or $a = -1$. **Tangent lines:** $a = 3$: $y = 6x - 9$; $a = -1$: $y = -2x - 1$.

14. Let x be the length of each of the two sides perpendicular to the river. The side parallel to the river is $120 - 2x$. Area $A(x) = x(120 - 2x) = 120x - 2x^2$, domain $0 < x < 60$. $A'(x) = 120 - 4x = 0 \iff x = 30$; A' changes from $+$ to $-$: maximum. **Dimensions:** 30 m by 60 m (the 60 m side along the river). **Area:** 1800 m².

15. $P = (x, \sqrt{x})$ with $x \geq 0$. $AP^2 = d(x) = (x-4)^2 + (\sqrt{x})^2 = x^2 - 7x + 16$. $d'(x) = 2x - 7 = 0 \iff x = \frac{7}{2}$; d is an upward parabola, so this is the minimum. $d\left(\frac{7}{2}\right) = \frac{49}{4} - \frac{49}{2} + 16 = \frac{15}{4}$. Since the square root is increasing, the distance is smallest where its square is smallest. **Smallest distance:** $\sqrt{\frac{15}{4}} = \frac{1}{2}\sqrt{15}$.

16a. $f'(x) = 3x^2 - 6x$, $f''(x) = 6x - 6$, which changes sign at $x = 1$. $f(1) = 1 - 3 + 4 = 2$. **Inflection point** $(1, 2)$.

16b. $f'(1) = 3 - 6 = -3$. $y = -3x + b$ through $(1, 2)$: $b = 5$. **Buigraaklijn:** $y = -3x + 5$.

16c. Intersect: $x^3 - 3x^2 + 4 = -3x + 5 \iff x^3 - 3x^2 + 3x - 1 = 0 \iff (x-1)^3 = 0 \iff x = 1$. So the only common point is the inflection point itself. □

17. $f(a) = \frac{1}{a}$, $f'(x) = -\frac{1}{x^2}$, $f'(a) = -\frac{1}{a^2}$. Tangent: $y = -\dfrac{1}{a^2}(x-a) + \dfrac{1}{a} = -\dfrac{x}{a^2} + \dfrac{2}{a}$. $x = 0$: $y = \frac{2}{a}$, so $B = \left(0, \frac{2}{a}\right)$. $y = 0$: $x = 2a$, so $A = (2a, 0)$. Area of the right triangle OAB: $\frac{1}{2} \cdot 2a \cdot \frac{2}{a} = 2$, independent of a. □

18a. Volume $\pi r^2 h = 500$, so $h = \dfrac{500}{\pi r^2}$. Surface: two discs and the side: $S = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \cdot \dfrac{500}{\pi r^2} = 2\pi r^2 + \dfrac{1000}{r}$.

18b. $S'(r) = 4\pi r - \dfrac{1000}{r^2} = 0 \iff r^3 = \dfrac{250}{\pi} \iff r = \sqrt[3]{\dfrac{250}{\pi}} \approx 4.30$ cm. For smaller r, $S' < 0$; for larger r, $S' > 0$: minimum. Height: $h = \dfrac{500}{\pi r^2} = \dfrac{2 \cdot 250}{\pi r^2} = \dfrac{2\pi r^3}{\pi r^2} = 2r$ (using $250 = \pi r^3$). So the height equals the diameter, ≈ 8.60 cm.

19a. $f''(x) = 12x^2$, so $f''(0) = 0$. But $f''(x) > 0$ for all $x \neq 0$, so f'' does not change sign at 0; the graph bends upwards on both sides. No inflection point (in fact $(0, 0)$ is a minimum).

19b. $g'(x) = 5x^4 - 20x^3$, $g''(x) = 20x^3 - 60x^2 = 20x^2(x-3)$. Zeros $x = 0$ and $x = 3$. At $x = 0$, $20x^2 \geq 0$ and $x - 3 < 0$ on both sides: no sign change. At $x = 3$, g'' changes from $-$ to $+$. $g(3) = 243 - 405 = -162$. **Inflection point:** $(3, -162)$ only.

20a. f' has zeros at $x = 1$ and $x = -2$. Sign: $(x-1)^2 \geq 0$, so the sign of f' is that of $x + 2$ (except at $x = 1$ where it is 0). At $x = -2$: from $-$ to $+$: **minimum**. At $x = 1$: f' is positive on both sides: **no extreme**.

20b. Expand: $f'(x) = (x^2 - 2x + 1)(x + 2) = x^3 - 3x + 2$. So $f''(x) = 3x^2 - 3 = 3(x-1)(x+1)$, which changes sign at $x = -1$ and $x = 1$. Inflection points at $x = -1$ and $x = 1$. (At $x = 1$ the graph has a horizontal inflection point, *horizontaal buigpunt*.)

Challenge

21a. *Base case* $n = 1$: $(x^1)' = 1 = 1 \cdot x^0$. *Step:* assume $(x^n)' = nx^{n-1}$. Then by the product rule $(x^{n+1})' = (x^n \cdot x)' = nx^{n-1} \cdot x + x^n \cdot 1 = (n+1)x^n$. So the formula holds for $n + 1$. By induction it holds for all positive integers. □

21b. $x^{-n} = \dfrac{1}{x^n}$. Quotient rule with numerator 1 (derivative 0): $\left(\dfrac{1}{x^n}\right)' = \dfrac{0 \cdot x^n - 1 \cdot nx^{n-1}}{x^{2n}} = -nx^{n-1-2n} = -nx^{-n-1}$. □

22a. $\dfrac{1}{h}\left(\dfrac{1}{g(x+h)} - \dfrac{1}{g(x)}\right) = \dfrac{g(x) - g(x+h)}{h\, g(x+h) g(x)} = -\dfrac{g(x+h) - g(x)}{h} \cdot \dfrac{1}{g(x+h)g(x)}$. As $h \to 0$: the first factor tends to $g'(x)$; $g(x+h) \to g(x)$ because a differentiable function is continuous. So the limit is $-\dfrac{g'(x)}{g(x)^2}$. □

22b. $\dfrac{f}{g} = f \cdot \dfrac{1}{g}$. Product rule and (a): $\left(\dfrac{f}{g}\right)' = f' \cdot \dfrac{1}{g} + f \cdot \left(-\dfrac{g'}{g^2}\right) = \dfrac{f'g}{g^2} - \dfrac{fg'}{g^2} = \dfrac{f'g - fg'}{g^2}$. □

23. $f''(x) = 6ax + 2b$, which is zero and changes sign at $x_0 = -\dfrac{b}{3a}$ (since $a \neq 0$). Point symmetry in $(x_0, f(x_0))$ means: $f(x_0 + h) + f(x_0 - h) = 2f(x_0)$ for all h. Compute $f(x_0 + h) + f(x_0 - h)$ term by term. Terms with odd powers of h cancel:

- $a((x_0+h)^3+(x_0-h)^3) = a(2x_0^3+6x_0h^2)$
- $b((x_0+h)^2+(x_0-h)^2) = b(2x_0^2+2h^2)$
- $c((x_0+h)+(x_0-h)) = 2cx_0$
- $d+d = 2d$

Sum: $2(ax_0^3+bx_0^2+cx_0+d)+h^2(6ax_0+2b) = 2f(x_0)+h^2\cdot f''(x_0) = 2f(x_0)$, since $f''(x_0)=0$. □

24a. $f'(a)=3a^2$, $f(a)=a^3$. Tangent: $y = 3a^2(x-a)+a^3 = 3a^2x-2a^3$.

24b. Intersect: $x^3-3a^2x+2a^3=0$. We know $x=a$ is a double root (the line touches there), so $(x-a)^2 = x^2-2ax+a^2$ is a factor. Division gives $x^3-3a^2x+2a^3 = (x-a)^2(x+2a)$. (Check: $(x^2-2ax+a^2)(x+2a) = x^3+2ax^2-2ax^2-4a^2x+a^2x+2a^3 = x^3-3a^2x+2a^3$.) So the other intersection is at $x=-2a$, which differs from a because $a\neq 0$. □

25. $f'(x)=3x^2+2ax+b$, $f''(x)=6x+2a$. Inflection point at $x=1$: $f''(1)=0 \Rightarrow a=-3$. (Then f'' is linear with a sign change at 1.) Extreme at $x=-1$: $f'(-1)=3-2a+b=3+6+b=0 \Rightarrow b=-9$. Check: $f'(x)=3x^2-6x-9=3(x-3)(x+1)$ changes from $+$ to $-$ at $x=-1$: a maximum. (Or $f''(-1)=-12<0$.) **Answer:** $a=-3$, $b=-9$.

26.

$$\frac{\sqrt{x+h}-\sqrt{x}}{h} = \frac{(\sqrt{x+h}-\sqrt{x})(\sqrt{x+h}+\sqrt{x})}{h(\sqrt{x+h}+\sqrt{x})} = \frac{(x+h)-x}{h(\sqrt{x+h}+\sqrt{x})} = \frac{1}{\sqrt{x+h}+\sqrt{x}}.$$

As $h\to 0$, $\sqrt{x+h}\to\sqrt{x}$, so the limit is $\dfrac{1}{2\sqrt{x}}$. □ (This agrees with the power rule: $\left(x^{1/2}\right)' = \frac{1}{2}x^{-1/2}$.)